

## An improved dual porosity model for chemical transport in macroporous soils

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### Abstract

The often observed processes involved in preferential water flow and chemical transport in porous media appear to be realistically described using a dual-continuum (dual-porosity) approach. In this approach, the porous medium is conceptualized as two coexistent continua, one representing the bulk matrix and the other the macropore region. Fluid and solute mass transfer between the two regions in the conceptual model occurs under pressure and concentration gradients. However, oscillatory behavior (overshoot problems in the macropore region) of the transport equation was observed for high values of the advective solute flux relative to the diffusive solute flux between the two regions. To circumvent this oscillatory behavior, the fluid coupling term in the transport equations was treated as an element-averaged, rather than a nodal property. The model was extended to two space dimensions for evaluating the impact of agricultural practices on solute leaching. A linear kinetic sorption module in the transport equations and a simple plant root extraction routine in the flow equations were also added. Although the simulation results show promise, additional work will be needed to determine realistic model parameter values.

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### 1. Introduction

Ground-water contamination from agricultural chemicals at national, state, and regional levels has been documented in a number of studies (Holden and Graham, 1990; Schock et al., 1992; US Environmental Protection Agency, 1992). When used as labelled, agricultural chemicals should not leach, according to regulatory screening models. Although

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measured rates of pesticide fate processes such as sorption/desorption, volatilization, and biodegradation suggest that these chemicals are relatively immobile, their appearance in surface and ground waters suggests otherwise. It is clear that the unsaturated zone plays an important role in controlling the arrival of these chemicals in ground water, in particular that the processes occurring in the unsaturated zone are not adequately characterized in these regulatory screening models, which are based on a conventional single continuum concept.

Numerous laboratory and field studies have pointed out that porous medium heterogeneity plays an important role in the preferential movement of water and solutes. Much of this work has been summarized in recent literature (e.g. Edwards et al., 1990; Pennell et al., 1990; Harrison et al., 1992; Gerke and van Genuchten, 1993a; Hutson and Wagenet, 1995; Mills et al., 1991; Gwo et al., 1995). Heterogeneity in the unsaturated zone arises from soil formation factors, biota, fractures, cracks, fissures, worm holes, large-scale textural variations, and inter-aggregate pore spaces. Soils exhibiting these features are frequently termed as “macroporous”. These porous media heterogeneities (hereafter referred to as “macropore” phenomena) lead to physical and chemical nonequilibrium (at a local scale) during water flow and chemical movement in soils. Macropores produce nonuniform flows with widely varying velocities. The conventional approaches to modelling flow and transport are not typically valid in such media.

A theoretical argument for the need of dual- or multi-continuum models is presented by McCoy et al. (1994). They discuss the phenomenon of “short-circuit bypass” flow in macroporous media: water in large pores bypassing dry soil in addition to bypassing slow-moving water. When this occurs, there will be small, relatively dry pores in the vicinity of large pores which contain water. Thus the phenomenon of “short-circuit bypass” violates the basic theory of soil-water retention. To accommodate this process to a single continuum system, a new water retention theory which includes the dynamic nonequilibrium between large and small pores at a local scale must be presented. Since such a theory is lacking at present, they suggest that “short-circuit bypass” phenomena can be described by a dual- or multi-continuum approach.

The dual-porosity modelling approach used in Gerke and van Genuchten (1993a) and Ray (1994) assumes Richards’ equation governs flow in both the matrix and the macropore region and also that the convection-dispersion equation governs solute transport within each of the two pore regions. Water and solute transfers between the two regions are represented by semi-empirical first-order rate expressions. In a numerical framework, two equations, one representing the matrix region and the other representing the macropore region, are simultaneously solved, while accounting for the transfer of water and solute from one region to the other.

We note that flow and transport in macroporous soils may be more completely described using multiregion models such as those presented by Gwo (1992), Gwo et al. (1995), Hutson and Wagenet (1995), and Morisawa et al. (1986 (cited by Brusseau and Rao, 1990)). While a multi-region approach may describe (fit) the field-scale processes more closely than a single or two-region approach, this is achieved at the cost of requiring a greater number of parameters for model calibration and raises concern about the physical relevance and significance of model parameters. Furthermore, incorporating variability in both soil physical and chemical properties would make the model extremely complex, and it is not yet clear that such complexity is experimentally justified.

Therefore, we have taken a dual-porosity approach as described in Gerke and van Genuchten (1993a) and Ray (1994) as the conceptual framework for developing a computer model to describe water flow and reactive chemical transport in macroporous media. The objective of the present work is to evaluate different formulations of the numerical solution and to assess the performance of the model for simulating flow and transport in macroporous soils under various agricultural management practices. The latter is achieved by studying the model under different rates of water and solute transfer between the two regions, using spatially-variable macropore fractions, rate limited-sorption processes, root water extraction, and first-order decay processes for simulating one- and two-dimensional flow and transport.

## 2. Conceptual modelling approach

### 2.1. The flow equations

The mass-conservative mixed form of the unsaturated flow equation, as proposed by Celia et al. (1990), is used to describe flow in each of the two regions. With the Einstein summing convention and  $x_3$  positive downward, the flow equation for each of the regions (matrix and macropore) is given by:

$$\frac{\partial(w_r\theta_r)}{\partial t} = \frac{\partial}{\partial x_i} \left[ w_r K_r(h_r)_{ij} \frac{\partial(h_r - x_3)}{\partial x_j} \right] \pm \Gamma_w - S_r w_r \quad (1)$$

where  $r$  equals  $m$  for the matrix region and  $f$  for the macropore region.  $\Gamma_w$  is a coupling term, defined as the volume of fluid moving from the macropore to the matrix region per unit bulk volume of the medium per unit time [1/T]. The  $\pm$  sign preceding  $\Gamma_w$  is  $+$  for  $r = m$  and  $-$  for  $r = f$ .  $K_r(h)_{ij}$  is the unsaturated hydraulic conductivity tensor [L/T] which is dependent upon the pressure head ( $h_r$ ), and  $S_r$  is a sink term which accounts, e.g. for root water uptake (volume of water uptake per unit volume of region per unit time) [1/T]. The term  $w_r$  is the matrix or the macropore volumetric fraction per unit volume of the bulk soil, with  $w_m + w_f = 1$  and  $\theta_r$  is the volume of water in the respective pore region per unit volume of that region.

Physical and chemical properties such as water content, hydraulic conductivity, concentration, and the sink terms can differ between the two regions. However, what is measured in the field is a composite of the two regions. For example, the following equation relates the individual region water contents to the composite:

$$\theta_{\text{comp}} = \theta_f w_f + \theta_m w_m \quad (2)$$

where the subscript comp refers to the composite property. The same principle applies for calculating composite fluxes, water capacities, and concentrations.

### 2.2. The flow coupling term

The coupling term is assumed to be proportional to the difference in pressure between the matrix and the macropore regions. In the absence of controlled laboratory or field

studies, it is difficult to predict a priori how the water transfer between the two regions is related to the properties in the two regions. So, the conceptual representation of the coupling term has been derived primarily from the literature on flow and transport in fractured rocks. Barenblatt et al. (1960) assumed that the coupling term is proportional to the pressure difference between the two regions. The constant of proportionality, an empirical fluid mass transfer coefficient, was assumed to be equal to the product of the matrix porosity and the square of the specific surface area of fractures. Duguid and Lee (1977) suggest that this relationship has no theoretical basis; the representation used by Duguid and Abel (1974) is based on a length scale associated with the size of the matrix block plus one-half of the fracture width, a shape factor for the matrix block, and fluid properties.

Othmer et al. (1991) also assumed that fluid coupling is proportional to the pressure difference between the two regions in an aggregated soil and further that their empirical fluid mass exchange term is equivalent to the minimum of the hydraulic conductivities of the matrix and the macropore regions divided by the sum of the radii of the average sized aggregates in the two regions. Gerke and van Genuchten (1993a) use the concept of Barenblatt et al. (1960) in defining the exchange term  $\Gamma_w$  [1/T], but with the fluid mass transfer coefficient a product of an empirical fluid mass exchange factor ( $\alpha_w$ ) and an unsaturated hydraulic conductivity at the hypothetical interface ( $K_a$ ). They defined the coupling term as:

$$\Gamma_w = \alpha_w K_a (h_f - h_m) \quad (3)$$

where  $\alpha_w = (\beta \gamma_w)/a^2$  with units [1/L<sup>2</sup>] is based on  $\beta$ , a geometry factor,  $a$ , the distance [L] from the center of a fictitious matrix block to the fracture boundary (equivalent to aggregate radius), and  $\gamma_w$ , an empirical coefficient that was estimated to be 0.4 by Gerke and van Genuchten (1993b). The geometry factor,  $\beta$ , equals 3 for rectangular slabs and 15 for spheres (see van Genuchten and Dalton, 1986).

Because of the complexity and uncertainty involved in obtaining these parameters, the exchange coefficient,  $\alpha_w$  in Eq. (3) is treated as an empirical constant in this work and is simply termed the coupling parameter. For a very large  $\alpha_w$ , the two regions reach nearly an instantaneous equilibrium, and for  $\alpha_w$  equal to zero the two regions behave independently. For significant preferential flow to occur, the coupling parameter must be small.

### 2.3. Transport equations

Two solute transport equations, one for each region, represent convective–dispersive transport, with rate limited sorption/desorption and first-order decay processes. Solute transfer between the two regions occurs as mass flow owing to water transfers and as diffusion, proportional to the concentration difference. In contrast to the nonlinear flow equations, the transport equations are linear. At each time step, after solution of the flow equations, the transport equations are solved using the calculated velocity, water content, and fluid exchange terms.

The solute transport equations are:

$$\frac{\partial(w_r \theta_r C_r)}{\partial t} + \frac{\partial(w_r \rho_{br} S_r)}{\partial t} = \frac{\partial}{\partial x_i} \left[ w_r \theta_r D_{r_{ij}} \frac{\partial C_r}{\partial x_j} - w_r q_r C_r \right] - w_r \theta_r \lambda_{1r} C_r - w_r \rho_{br} \lambda_{2r} S_r \pm \Gamma_s \quad (4)$$

where the subscript  $r$  is  $m$  for the matrix region and  $f$  for the macropore region,  $C_r$  is the concentration of a constituent in the liquid phase [ $M/L^3$ ],  $S_r$  is its concentration in the solid phase [ $M/M$ ],  $D_r$  is the dispersion coefficient [ $L^2/T$ ],  $q_r$  is the Darcy velocity [ $L/T$ ],  $\rho_{br}$  is the bulk density [ $M/L^3$ ],  $\lambda_{1r}$  and  $\lambda_{2r}$  are first-order decay coefficients [ $1/T$ ] (Valentine and Schnoor, 1986; van Genuchten, 1987) for the liquid and the solid phases, respectively, and  $\Gamma_s$  is the solute coupling term [ $M/(L^3T)$ ].

The solute coupling term  $\Gamma_s$  is defined as the mass flux of solute from one region to another per unit time per unit bulk volume of the soil and is taken to be:

$$\Gamma_s = (1-d)\Gamma_w C_f + d\Gamma_w C_m + \alpha_s \Theta_{mf} (C_f - C_m) \quad (5)$$

where  $d$  is a direction switch. If  $\Gamma_w$  is positive (i.e. for flow from the macropore to the matrix region),  $d$  is equal to zero, and for  $\Gamma_w$  negative,  $d = 1$ . Thus the first two terms on the right in Eq. (5) define the advective exchange of solute owing to the movement of water from one region to the other under the water pressure gradient between the two regions. The last term in Eq. (5) accounts for the diffusive exchange of solute owing to a concentration gradient with  $\alpha_s$  an empirical solute exchange coefficient between the two regions and where we define  $\Theta_{mf} = \theta_m \theta_f \kappa$ , with  $\kappa$  a coefficient whose value is unity with units so that  $\Theta_{mf}$  has units of volume of water per unit volume of porous medium; in this way  $\alpha_s$  has units of [ $1/T$ ].

The concept of  $\alpha_s$  in the above equations follows from that developed for solute transport under steady-state flow in two-region models (van Genuchten and Dalton, 1986; Parker and Valocchi, 1986; Valocchi, 1990). Gerke and van Genuchten (1993a) used  $\alpha_s = (\beta D_a/a^2)$ , where  $D_a$  is an effective molecular diffusion coefficient [ $L^2/T$ ] of the matrix block near the interface (which is analogous to  $K_a$  in Eq. (3)) and  $\beta$  and  $a$  are as defined earlier. Since the physical significance of  $\beta$  and  $a$  are questionable and the ability to measure them is uncertain, in our model we assumed, contrary to Gerke and van Genuchten (1993a), that  $\alpha_s$  is an empirical solute exchange coefficient.

The sorption/desorption process indicated by the second term on the left in Eq. (4) is assumed to follow a first-order reversible rate law (Boast, 1973; Valocchi, 1989):

$$\frac{\partial(w_r \rho_{br} S_r)}{\partial t} = w_r k_r (K_{dr} \theta_r C_r - \rho_{br} S_r) \quad (6)$$

where  $r$  is as defined following Eq. (1),  $k_r$  is the reaction rate constant [ $1/T$ ] and  $K_{dr}$  is the distribution coefficient. When the reaction rate is very fast, i.e. when  $k_r$  is very large, the above equation reduces to the widely used linear equilibrium isotherm.

### 3. Numerical solution: flow

We have used two separate formulations of the flow and transport equations and developed a computer code for each. The first code is for one-dimensional flow and transport in the vertical direction, and the other code is for two-dimensional flow and transport in the vertical plane. A significant difference between the two formulations is that the one-dimensional model assumes an element-wise constant velocity, whereas the two-dimensional code assumes the velocity to vary linearly over the element. In addition,

the codes differ with respect to computing the value of the interfacial conductivities,  $K_a$ . Computation of the interfacial conductivity in the two-dimensional model depends upon the matrix conductivity in the  $X$  and the  $Z$  directions and is discussed in detail in Ray (1994). In both formulations, the computation of the root water extraction term (the sink term ( $S_r w_r$ )) follows that given in van Genuchten (1987).

The Galerkin finite element method with linear shape functions (e.g. Istok, 1989; Celia and Gray, 1992) is utilized for the numerical solution of the flow equation in each region. Eq. (1) is a mixed-form Richards' equation with the unknown variables expressed in terms of  $\theta$  and  $h$ . The modified Picard method suggested by Celia et al. (1990) is used to express the two unknowns in terms of changes in pressure head ( $\delta h$ ) between iterations. The backward Euler method is used to discretize the time derivative terms on the left sides of these equations and the general Galerkin technique is used for the space discretization. The soil water capacity term,  $C (= d\theta/dh)$ , is evaluated analytically.

We substitute the basis function expansion for the unknown variables into a difference form of Eq. (1) for the two regions. Integration by parts is then used to reduce the order of the derivative terms. We also mass lump the time derivative terms. These operations are explained in detail in many texts (e.g. Celia and Gray, 1992; Istok, 1989). We group the terms  $w_r C_r$  and  $w_r K_r$  before integration. The discretized form of the resulting equation for the matrix and the macropore region produces the following linear system:

$$[A1_m]\{\delta h_m\}^{n+1,k} + [A2_m]\{\delta h_f\}^{n+1,k} = [A3_m]\{h_m\}^{n+1,k} + [A4_m]\{h_f\}^{n+1,k} + \{g1_m\} \quad (7)$$

$$[B1_f]\{\delta h_f\}^{n+1,k} + [B2_f]\{\delta h_m\}^{n+1,k} = [B3_f]\{h_f\}^{n+1,k} + [B4_f]\{h_m\}^{n+1,k} + \{g1_f\} \quad (8)$$

where  $\delta h^{n+1,k} = h^{n+1,k+1} - h^{n+1,k}$ , the  $A$ s and  $B$ s are coefficient matrices, and the  $g$ s are the known vectors. The explicit forms of these coefficient matrices are in Ray (1994). The superscript  $n$  refers to the time step and  $k$  to the iteration step. At the start of the solution we normally guess  $h^{n+1,0}$  to be same as  $h_n$  (which is the initial value). For the  $k$ th iteration ( $k = 1, 2, \dots$ ),  $h^{n+1,k}$  is known and we compute  $\delta h^{n+1,k}$  and hence  $h^{n+1,k+1}$  at the end of the iteration. This  $h^{n+1,k+1}$  serves as  $h^{n+1,k}$  in the next iteration. This is repeated until  $\delta h^{n+1,k}$  becomes smaller than a set tolerance and at the same time, the right sides of Eqs. (7) and (8) become smaller than a threshold value.

#### 4. Numerical solution: transport

As for flow, two codes are available for one- and two-dimensional transport problems. Unlike flow codes, the discretization and integration techniques are the same for the one- and two-dimensional models. The first term in Eq. (4) is expressed using simple finite differences.

$$\frac{\partial(w_r \theta_r C_r)}{\partial t} = \frac{(w_r \theta_r C_r)^{n+1} - (w_r \theta_r C_r)^n}{\Delta t} \quad (9)$$

where the superscripts  $n$  and  $n + 1$  refer to the old and the new time levels, respectively. We followed a variably-weighted Euler time stepping (Celia, 1992) to discretize all other terms in the pair of transport equations. For the time weighting factor  $\epsilon = 1.0$ , the scheme is fully implicit, and for  $\epsilon = 0.5$  it is the Crank–Nicholson scheme. The time discretization of Eq. (6) using the variably weighted scheme is:

$$\frac{\rho_{b_r} S_r^{n+1} - \rho_{b_r} S_r^n}{\Delta t} = \epsilon [k_r (K_{d_r} \theta_r C_r - \rho_{b_r} S_r)]^{n+1} + (1 - \epsilon) [k_r (K_{d_r} \theta_r C_r - \rho_{b_r} S_r)]^n \quad (10)$$

Solving Eq. (10) and substituting this into Eq. (4) gives the unknowns in terms of  $C_r^{n+1}$  only.

The fully implicit equation ( $\epsilon = 1.0$ ) for the matrix region in the one-dimensional system illustrates the variables involved:

$$\begin{aligned} & \left\{ \frac{w_m \theta_m}{\Delta t} + \frac{k_m K_{d_m} \theta_m w_m}{1 + k_m \Delta t} + \frac{\lambda_{2_m} \Delta t (k_m K_{d_m} \theta_m w_m)}{1 + k_m \Delta t} + \lambda_{1_m} \theta_m w_m \right\}^{n+1} C_m^{n+1} \\ & - [d\Gamma_w - \alpha_s \theta_m \theta_f \kappa]^{n+1} C_m^{n+1} - \frac{\partial}{\partial z} \left[ w_m \theta_m D_m \frac{\partial C_m}{\partial z} - w_m q_m C_m \right]^{n+1} \\ & - [(1 - d)\Gamma_w + \alpha_s \theta_m \theta_f \kappa]^{n+1} C_f^{n+1} = \left\{ \frac{w_m \theta_m}{\Delta t} \right\}^n C_m^n + \left( \frac{\rho_{b_m} w_m k_{f_m}}{1 + k_m \Delta t} \right)^n S_m^n \\ & - \left\{ \frac{\lambda_{2_m} \rho_{b_m} w_m}{1 + k_m \Delta t} \right\}^n S_m^n \end{aligned} \quad (11)$$

In Eq. (11), the variables on the left side of the equation are at the new time-level (superscript  $n + 1$ ) and the variables on the right side are at the old time level (superscript  $n$ ), and  $d$  is as described following Eq. (5). The standard Galerkin approach is taken and all time-derivative terms are mass lumped (Istok, 1989). Terms such as  $w_m \theta_m$ ,  $\theta_m \theta_f$ , and  $w_m \theta_m D_m$  are grouped together before integration. The term  $w_m \theta_m D_m$  in the dispersive flux is calculated as the sum of a dispersion coefficient and a molecular diffusion component for the one-dimensional model at the center of an element (element averaged) using the equation:

$$(\overline{w_m})(\overline{\theta_m})D_m = \overline{w_m} |q_m| \alpha_{l_m} + (\overline{w_m})(\overline{\theta_m})D_{mol} \quad (12)$$

where  $\overline{\theta}$  is the arithmetic average water content over the element (using the two nodal values),  $w_m$  is the arithmetic average of matrix volume fractions over the element,  $\alpha_l$  is the longitudinal dispersivity [L] of the medium (which is constant for a given material type),  $|q|$  is the absolute value of the element-averaged Darcy velocity, and  $D_{mol}$  is the molecular diffusion coefficient (constant for a given material type). We also assume that in the term  $(w_m q_m C_m)$  in the left side of Eq. (11), the advective flux  $q_m$  is constant over an element, while allowing  $w_m$  to be variable. Similar considerations apply for the macropore region.

The computation of dispersion coefficients for the two-dimensional version of Eq. (4) is made using the convention of Bear (1972) (see also Simunek et al., 1992)

$$\theta D_{ij} = \alpha_T |q| \delta_{ij} + (\alpha_L - \alpha_T) \frac{q_i q_j}{|q|} + \theta D_{mol} \delta_{ij} \quad (13)$$

where  $\alpha_L$  and  $\alpha_T$  are the longitudinal and transverse dispersivities [L], respectively, and  $\delta_{ij}$  is the Kronecker delta function ( $\delta_{ij} = 1$  for  $i = j$ , or else  $\delta_{ij} = 0$ ).

The fluid coupling term ( $\Gamma_w$ ) was integrated using a consistent formulation (Istok, 1989). The standard Galerkin technique with linear basis functions was used for the solution of the pair of transport equations giving a linear system of equations. After all the integrations were performed and rearranged, we obtained the following system (which is nonsymmetric) for the two regions:

$$[P1_m]\{C_m\}^{n+1} + [P2_m]\{C_f\}^{n+1} = [P3_m]\{C_m\}^n + [P4_m]\{C_f\}^n + [G_m]\{S_m\}^n + \{h_m\} \quad (14)$$

$$[P1_f]\{C_f\}^{n+1} + [P2_f]\{C_m\}^{n+1} = [P3_f]\{C_f\}^n + [P4_f]\{C_m\}^n + [G_f]\{S_f\}^n + \{h_f\} \quad (15)$$

where the square brackets represent the coefficient matrices and the vectors  $\{h_m\}$  and  $\{h_f\}$  represent the boundary terms for the matrix and the macropore regions, respectively. Explicit expressions for these coefficient matrices are presented in Ray (1994). After solving for  $C^{n+1}$ ,  $S^{n+1}$  is calculated using Eq. (10).

## 5. Computation of water content and Darcy velocity

Water content and Darcy velocity are used in the solution of the transport equations. The procedures used to compute the water content at a node within an element in the one- and the two-dimensional codes are the same and these calculations are based upon the pressure at the node and the material property of the element containing that node.

The procedure for explicit computation of water contents for the entire domain differs between the one- and the two-dimensional codes, and with heterogeneities of the medium. For the one-dimensional code in the presence of soil layers, we identify the nodes located at the layer interface and recompute the water content at those nodes first using the material property of the medium above it and again using the material property of the medium below it. The final water content is a weighted average of the two where the weighting factors are the lengths of the elements. The two-dimensional code internally stores the node-element connectivity information. For each element connecting a given node, we compute the water content using the material property of that element with the final water content at that node as an unweighted arithmetic average.

The Darcy velocity arises in the advective term in the transport equations. In the one-dimensional code, the computation of the hydraulic gradient applies to the mid-point of an element. The Darcy velocity was calculated using the computed hydraulic gradient and an arithmetic average of the nodal hydraulic conductivity values. As a result, the velocity was piecewise constant. In the two-dimensional code, nodal velocities for triangular and rectangular elements were obtained using the methods presented by Pinder et al. (1981) and Celia (1992). Details of these computations are presented in Ray (1994).



## 6. Numerical difficulties and model performance

### 6.1. One-dimensional model

Using the formulations of Gerke and van Genuchten (1993a), numerical overshoot problems were encountered in the macropore region. In their approach (which we initially followed), nodal values of  $\Gamma_w$  in the one-dimensional transport code were calculated using the nodal pressure head values for the matrix and the macropore regions, the interfacial conductivity at the node, and the fluid mass transfer coefficient, as shown in Eq. (3). Then  $\Gamma_w$  was expanded using linear basis functions and incorporated into the transport equations.

Code performance for this formulation is illustrated in a hypothetical soil column of depth 60 cm with a uniform initial pressure of  $-100$  cm. The material properties within each region were assumed to be uniform over the column and the volume fraction of the macropore region was set to either 2 or 10%. The column was discretized into 2.0 cm long elements. The unsaturated hydraulic properties for each region were represented using the Mualem–van Genuchten relationships (Mualem, 1976; van Genuchten, 1980):

$$\theta = \theta_r + (\theta_s - \theta_r)[1 + |\alpha h|^n]^{-m} \quad (16)$$

$$K = K_s S_e^{1/2} [1 - (1 - S_e^{1/m})^m]^2 \quad (17)$$

$$S_e = \frac{(\theta - \theta_r)}{(\theta_s - \theta_r)}, m = 1 - 1/n \quad (18)$$

with  $\theta_s$  the saturated water content,  $\theta_r$  the residual water content,  $K_s$  the saturated hydraulic conductivity [L/T],  $S_e$  the effective saturation, and  $\alpha$  and  $n$  as shape (or fitting) parameters. The inverse of  $\alpha$  in Eq. (16) is proportional to the air entry or bubbling pressure. Although there are numerous experimental observations of the  $\theta(h)$  and  $K(h)$  or  $K(\theta)$  relationships for soil viewed as a single continuum system (see especially van Genuchten et al., 1991), there are few experimental estimates of  $\theta(h)$  and  $K(\theta)$  for individual regions in a dual or multipore region model. Othmer et al. (1991) and Wilson et al. (1992) present schemes for obtaining individual region data from composite data. However, these composite data are based on equilibrium measurements, while it is the nonequilibrium process that must be observed to accurately estimate these properties for a dual-porosity soil. Lacking experimental data, we arbitrarily selected hydraulic properties such that the macropore region dominates flow close to saturation. These data are shown in Table 1.

For the simulation, the pressure at the top boundary was set to zero (i.e.  $h = 0$  at  $z = 0$ ) and the bottom was a unit hydraulic gradient ( $\partial h / \partial z = 1$  at  $z = L$ ). The total simulation time was 500 s using 5 s time steps. In the nonlinear Picard solution, the maximum number of iterations allowed per time step was 25 and the solution was assumed to have converged when the pressure change ( $\delta h$ ) between successive iterations was  $\delta h \leq 0.1$  cm. In addition, the residual pressure tolerance (right sides of Eqs. (7) and (8)) between iterations was set to  $1.0 \times 10^{-6}$  cm. If convergence required more than 25 iterations for a time step, the solution was restarted with the time step size cut in half.

The fluid interaction coefficient ( $\alpha_w$ ) was set to either 0, 0.5, 10, or  $1000 \text{ cm}^{-2}$ . The

Table 1

Unsaturated hydraulic properties used in simulation

| Type       | $\theta_r$ | $\theta_s$ | Exponent ( $n$ ) | $\alpha$ ( $\text{cm}^{-1}$ ) | $K_s$ ( $\text{cm s}^{-1}$ ) |
|------------|------------|------------|------------------|-------------------------------|------------------------------|
| Macropores | 0.102      | 0.3681     | 2.0              | 0.03353                       | 0.00922                      |
| Matrix     | 0.10       | 0.391      | 1.48             | 0.59                          | 0.0003639                    |

pressure profiles for the soil with 10% macroporosity are shown in Fig. 1 and the corresponding water contents in Fig. 2. Similarly, Figs 3 and 4 show the pressure and water content profiles, respectively, for the soil with 2% macropores. Fig. 1(a) and Fig. 3(a) are identical since the interaction coefficient is zero in each case and regions with identical hydraulic properties produce identical pressure profiles within those regions, irrespective of their relative volumes.

As can be seen from Figs. 2, and 4, especially from Fig. 2, the macroporosity wets the soil to a considerably greater depth than would have occurred in the case of single region with properties of the matrix region. Note that the penetration of the water front (in the macropore region) was retarded with increased  $\alpha_w$  indicating that for large coupling the two regions equilibrate quickly. The deviation of composite water content (what is measured in the field) from what one would expect using a single-region computer model in Figs. 2 and 4 and could easily be viewed as measurement error, if the effect of the macropores were not considered. Note also, that these figures, especially Fig. 2(a), (b), (c) illustrate the “short-circuit bypass” phenomena discussed by McCoy et al. (1994), i.e. water filled large pores at the same depth as relatively dry smaller pores.

The impact of the fluid mass transfer coefficient ( $\alpha_w$ ) on solute transport was studied in the same one-dimensional system. First, the solute mass transfer coefficient ( $\alpha_s$ ) and all reaction and decay coefficients were set to zero. This simulated nonreactive advection–dispersion in each region with advective solute exchange the only transfer mechanism between the two regions, i.e. no diffusive exchange. The input solute concentration at the

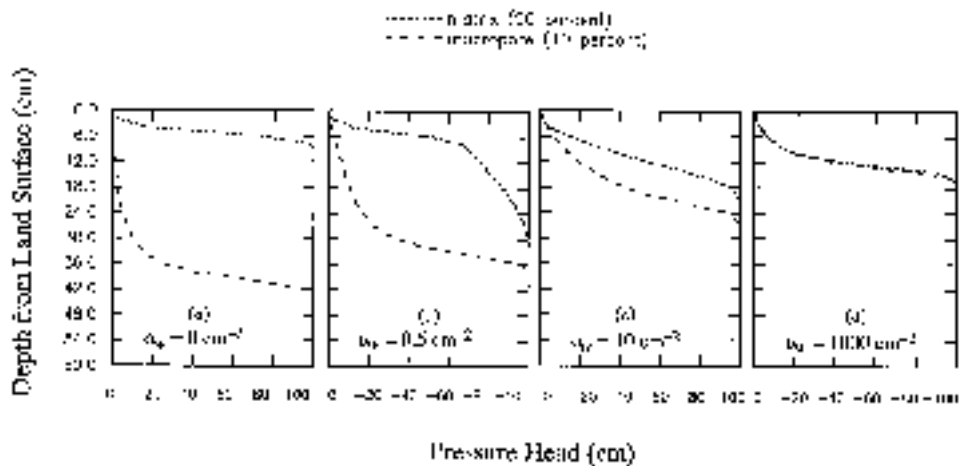


Fig. 1. Pressure head versus depth for a dual porosity soil with 10% macropores for various values of  $\alpha_w$ .

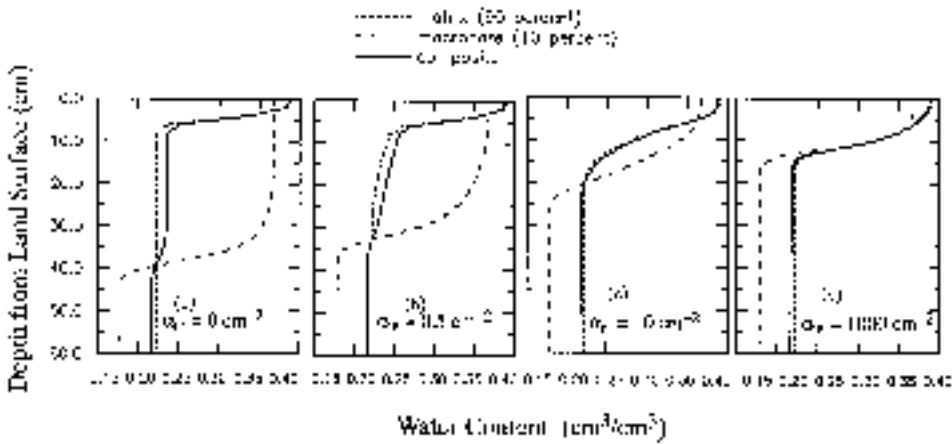


Fig. 2. Water content versus depth for a dual porosity soil with 10% macropores for various values of  $\alpha_w$ .

surface source was  $1.0 \text{ mg L}^{-1}$ , and in Eq. (13), the molecular diffusion coefficient was  $1.0 \times 10^{-5} \text{ cm}^2 \text{ s}^{-1}$ , and the longitudinal dispersivity was  $1.0 \text{ cm}$ .

The concentration profiles for 10% macroporosity are shown in Fig. 5 and for 2% macroporosity in Fig. 6. For both cases, as  $\alpha_w$  increased, the relative concentration in the macropore region increased to values greater than 1.0 a result which is physically impossible for the given input concentration. It is speculated that in the presence of strong advective solute exchange, a numerical instability in the coupled solution of the transport equation caused this behavior. This problem is addressed below.

6.2. Two-dimensional model

The two dimensional simulations were carried out for a soil block of cross-section  $38 \text{ cm} \times 48 \text{ cm}$  in the  $X$ – $Z$  directions, respectively. The domain was discretized into

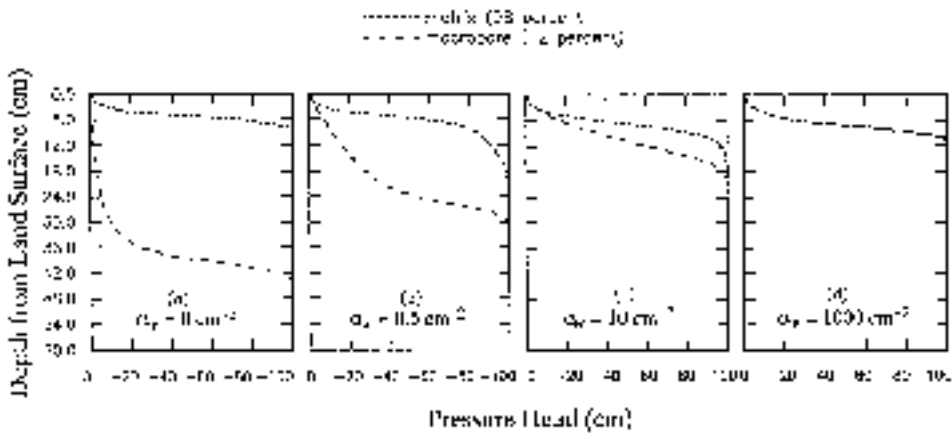


Fig. 3. Pressure head versus depth for a dual porosity soil with 2% macropores for various values of  $\alpha_w$ .

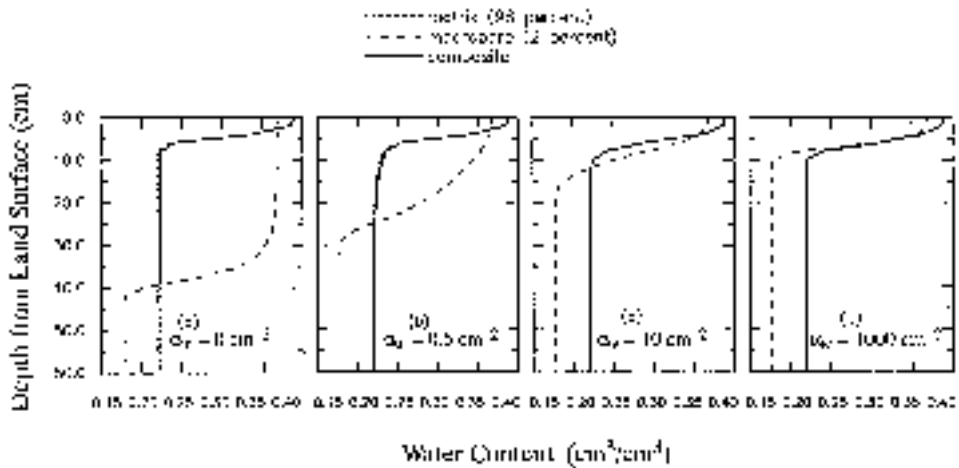


Fig. 4. Water content versus depth for a dual porosity soil with 2% macropores for various values of  $\alpha_w$ .

456 elements each  $2.0 \text{ cm} \times 2.0 \text{ cm}$ . Altogether, 500 nodes were present in the domain producing 1000 degrees of freedom for the coupled system. The soil hydraulic properties are the same as for the one-dimensional simulations. The initial pressure in the block was  $-100 \text{ cm}$  everywhere. A  $10 \text{ cm}$  wide, zero pressure horizontal strip source of infiltration was imposed on the upper left corner of the domain. At the bottom of the domain a Dirichlet boundary condition of  $-100 \text{ cm}$  pressure was imposed, and the remaining portions of the boundary had zero flux. A fixed time step size of  $1.0 \text{ s}$  was used in the simulations and the total simulation period was  $500 \text{ s}$ . The soil macropore fractions were the same as in the one-dimensional model, either 2 or 10%. Calculations were made for fluid mass transfer coefficient values ranging from 0 to  $200 \text{ cm}^{-2}$ . Pressure profiles for  $\alpha_w = 200 \text{ cm}^{-2}$  are shown in Fig. 7(a) and (b). There is greater water penetration in the macropore region than in the matrix region, and the vertical extent of the infiltrating water in both the matrix and the macropore regions is 40–60% greater in the

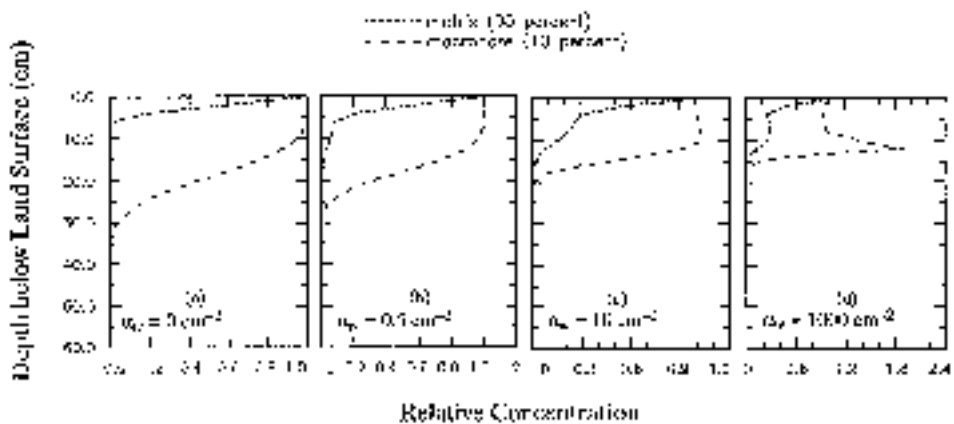


Fig. 5. Relative concentration versus depth for a dual porosity soil with 10% macropores for four values of  $\alpha_w$ .

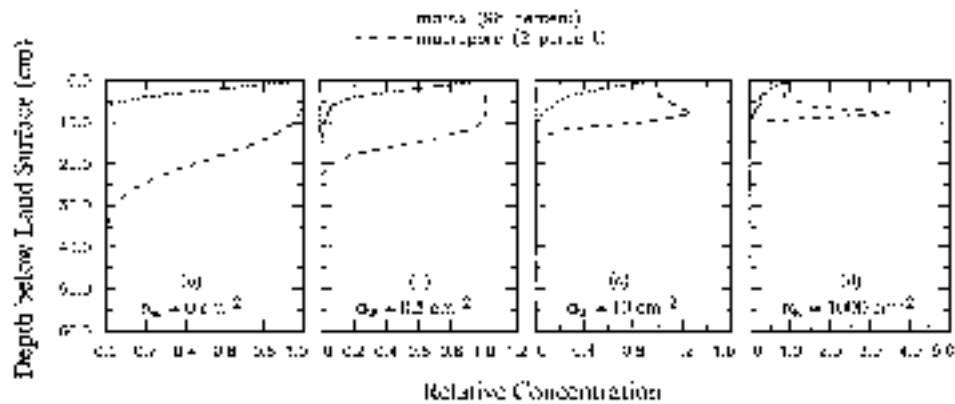


Fig. 6. Relative concentration versus depth for a dual porosity soil with 2% macropores for four values of  $\alpha_w$ .

soil with 10% macropores than in the soil with 2% macropores. The discrepancy in the horizontal penetration is less. For the matrix region the increased wetting in the soil with 10% macropores is a consequence of the greater total amount of water entering the soil through the macropore region.

Non-reactive transport was simulated for the same two-dimensional strip infiltrations. The composite soil system had a solute free initial condition. The surface boundary condition at the strip source was a Dirichlet type with unit concentration, the bottom boundary was zero concentration (which is appropriate for a short duration simulation, in which the front is far from the bottom boundary), and the remaining boundaries were zero solute flux boundaries. As in the one-dimensional case, simulations were carried out in the absence of diffusive solute transfer ( $\alpha_s = 0$ ). The concentration profiles after 500 s of simulation, presented in Fig. 7(c) and Fig. 7(d), exhibit oscillations. This phenomenon was also observed for smaller  $\alpha_w$  values, but to a lesser extent.

7. Identification of factors controlling oscillation and subsequent model improvements

Various approaches were taken to find the cause of the oscillations and overshoots in concentration profiles in the macropore region during simulations with large values of  $\alpha_w$ . As a first approach, the space and time discretizations were refined. For the element size reduced from 2.0 to 0.5 cm, with the time step size maintained at 5.0 s, concentration profiles at the end of 500 s are shown in Fig. 8. Comparing Figs. 5 and 6 with Fig. 8 shows that a four-fold finer space discretization removed much of the overshoot problem, especially for 10% macropores. Reduction of the time step size from 5.0 s to 1.0 s did not improve the concentration profiles for either spatial discretization.

Given the computational cost associated with a four-fold reduction in element size, especially for two-dimensional calculations, these observations led us to reexamine the finite element formulation of the solute coupling term ( $\Gamma_s$ ). As seen in Eq. (5),  $\Gamma_s$  includes an advective solute flux component, resulting from fluid flow between the two regions.

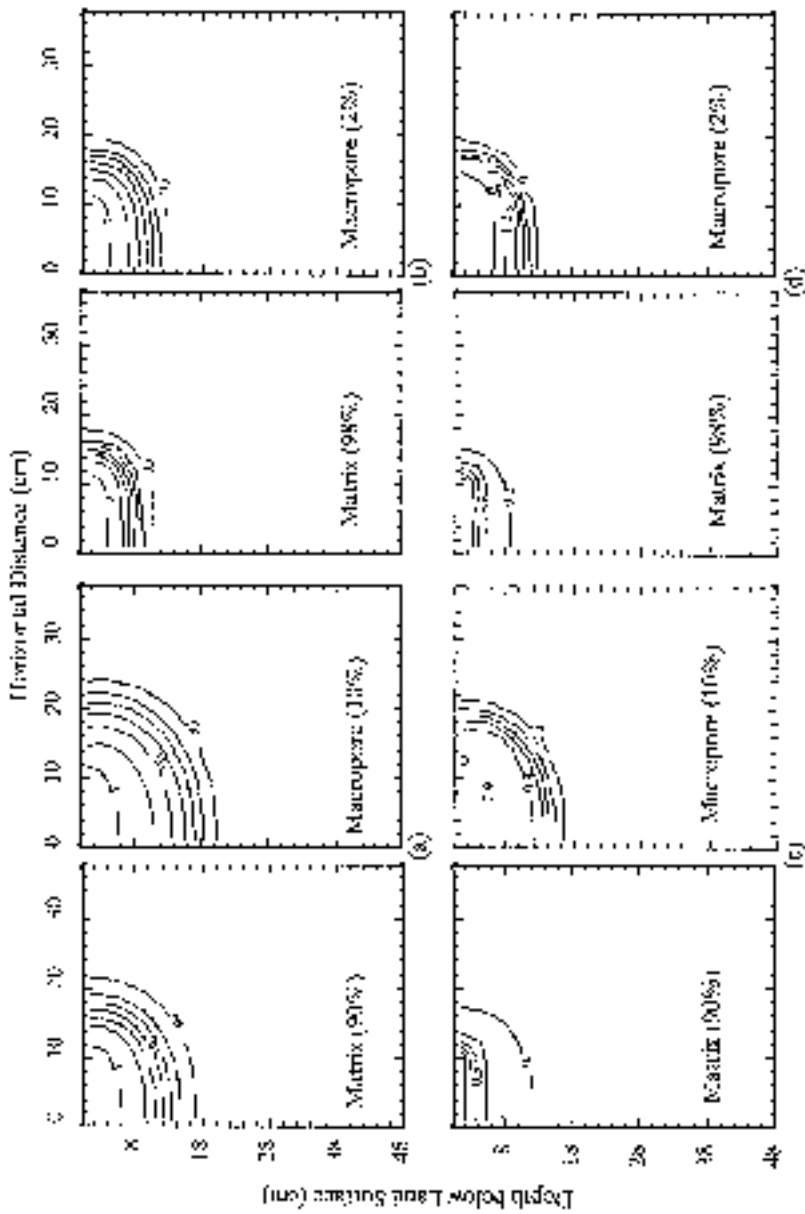


Fig. 7. Flow and transport in a dual porosity soil for a strip infiltration source with  $\alpha_w = 200 \text{ cm}^{-2}$ , the upper figures show pressure (cm) profiles and the lower figures show relative concentration profiles.

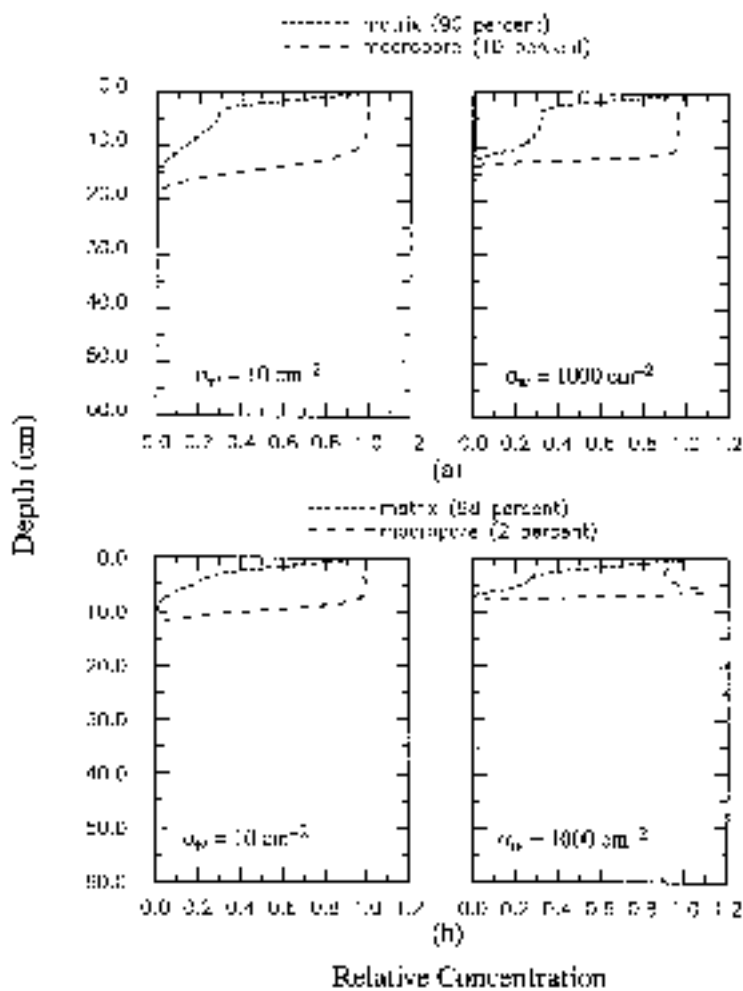


Fig. 8. Relative concentration versus depth for a dual porosity soil with (a) 10% and (b) 2% macropores for element sizes of 0.5 cm and time steps of 5 s.

Our initial Galerkin finite element procedure (and that used by Gerke and van Genuchten, 1993a) computes nodal values of  $\Gamma_w$  based upon a consistent formulation (Istok, 1989). Since  $\Gamma_w$  can be viewed as a rate expression, we chose to mass lump this term (diagonalize the entries of the element matrix for terms with  $\Gamma_w$ ), similar to the way in which the other time derivative terms in the solute mass balance equation are handled. Mass lumping had little impact on the problems shown in Fig. 5, and actually worsened behavior for  $\alpha_w = 10 \text{ cm}^2$  with 10% macropores.

Since the fluid flux in the one-dimensional problem is piecewise constant (see Section 4), we next computed an element averaged fluid coupling term ( $\overline{\Gamma_w^e}$ ) and used that as input in solving the transport equation. The computation of  $\overline{\Gamma_w^e}$  is made by

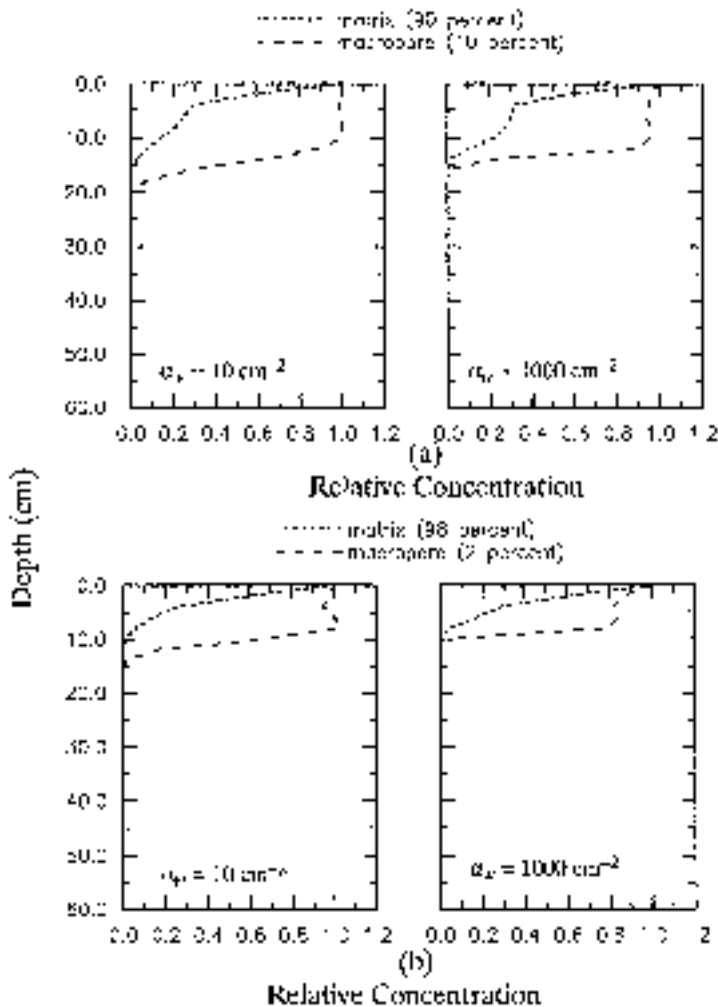


Fig. 9. Relative concentration versus depth for a dual porosity soil with (a) 10% and (b) 2% macropores for with element lengths of 2.0 cm and time steps of 5.0 s using a consistent formulation with element averaged fluid mass transfer coefficient.

integrating  $\Gamma_w$  over an element using the following formula

$$\overline{\Gamma_w^e} = \frac{1}{\Delta z} \int \Gamma_w dz = \frac{1}{\Delta z} \alpha_w \int K_a(h_f - h_m) dz \quad (19)$$

where  $\Delta z$  is the length of the element and the other terms are as defined previously. Each of the terms in the integrand is expanded using linear basis functions. Using Eq. (19),  $\overline{\Gamma_w}$  is then calculated for each element. Concentration profiles calculated using this strategy (Fig. 9) exhibit essentially no overshoot problems in the macropore region although non-overshoot oscillations are still present.



For comparison purposes, another approach was tried: nodal values of the fluid velocity were used in the solution of the transport equation. In this effort, the Darcy equation at each node was expanded using linear basis functions and the general Galerkin approach was again used. Since the coefficient matrix is time invariant, only the right side of Eq. (11) for the matrix and a similar equation for the macropore region needed to be computed in each time step. For this approach, two strategies for computing  $\Gamma_w$  were tried: (a) nodal values of  $\Gamma_w$  were computed, so as to be consistent with the nodal velocity and (b)  $\Gamma_w$  was computed as an element averaged value, as described in Eq. (19). Space and time discretizations of 2.0 cm and 5.0 s were used. As expected, the nodal computation of  $\Gamma_w$  produced overshoot problems comparable to those shown in Fig. 5. For element-averaged  $\Gamma_w$ , the overshoot problems were considerably reduced, although not as much as is shown in Fig. 9.

All the above simulations were performed subject to zero diffusive mass transfer ( $\alpha_s = 0$ ). Concentration simulations for a mass transfer coefficient which is based on a realistic molecular diffusion coefficient ( $\alpha_s$  of  $0.00005 \text{ s}^{-1}$ ) were carried out for cases in which  $\Gamma_w$  is computed nodally. The values of  $\alpha_w$  ranged from 0.5 to  $1000 \text{ cm}^{-2}$  and the same set of conditions as used previously were employed. No noticeable change in the numerical overshoot problem in the macropore was observed. However, a 100-fold increase in  $\alpha_s$  ( $0.005 \text{ s}^{-1}$ ) in this formulation considerably reduced the overshoot problem for all  $\alpha_w$  values. Increasing the value of  $\alpha_s$  beyond  $0.005 \text{ s}^{-1}$  gradually brought the concentration profiles in the two regions closer to each other.

As discussed earlier, the two-dimensional code using nodal velocities in the transport equation, and nodal values of  $\Gamma_w$  resulted in oscillations in concentration profiles in the macropore region for large values of  $\alpha_w$ . Therefore, we developed a version of the two-dimensional code for nodally computed velocity and element-averaged  $\Gamma_w$  which we found to effectively remove the oscillations. To conserve space, we do not reproduce these results from the improved two-dimensional code here.

## 8. Example application

### 8.1. Selection of soil and simulation parameters

The one- and the two-dimensional models were used to evaluate the impact of specific agricultural management practices on the preferential movement of water and reactive chemicals in the vadose zone. Since the model was not calibrated to specific field data, nor are estimates of the hydraulic and transport parameters for the individual regions of a composite porous media available, we chose parameter values thought to be representative of soils and management practices common in Illinois.

Properties of Drummer silty clay loam soil, typical for this area, were used to represent the matrix region in the simulations. Drummer is a poorly drained to moderately permeable soil found on outwash or till plains. The Soil Survey Manual (US Department of Agriculture, 1982) for Champaign County, IL, lists typical A and B horizon depths to be 35 and 119 cm, respectively, with many subhorizons. The composite saturated hydraulic conductivity ranges from  $4.2 \times 10^{-4}$  to  $8.5 \times 10^{-4} \text{ cm s}^{-1}$ . The bulk density for typical

Table 2

Hypothetical unsaturated hydraulic properties for macroporous Drummer soil

| Type      | $\theta_r$ | $\theta_s$ | Exponent ( $n$ ) | $\alpha$ (cm <sup>-1</sup> ) | $K_s$ (cm s <sup>-1</sup> ) |
|-----------|------------|------------|------------------|------------------------------|-----------------------------|
| Macropore | 0.045      | 0.43       | 2.68             | 0.145                        | 0.00825                     |
| Matrix    | 0.068      | 0.33       | 1.25             | 0.036                        | 0.00011944                  |

Drummer soils as measured by Kamau et al. (1996) ranged from 1.4 g cm<sup>-3</sup> at the land surface to 1.5 g cm<sup>-3</sup> at a depth of 100 cm. The organic matter content over the same depths decreased from 7.5 to 0.5% by volume (Keefer, 1993). The soil-water retention and hydraulic conductivity functions for the matrix region were selected from existing data for glacial materials (van Genuchten et al., 1991). The chosen macropore region properties correspond to those of a coarse sand with a saturated hydraulic conductivity nearly 70 times greater than for the matrix region. The saturated hydraulic conductivity of the interface was set equal to 0.01 times the saturated hydraulic conductivity of the matrix following the approach of van Genuchten and Gerke (1992). The unsaturated hydraulic properties of the two regions are given in Table 2.

We assume that the fraction of macroporosity ( $w_f$ ) varied with depth and also that this variation was proportional to the variation in soil aggregate stability in the A and B soil horizons. We used a relationship given by Heinonen (1985) to estimate aggregate stability using the percentage of organic matter content, and then we used this aggregate stability depth profile to estimate the variability of  $w_f$ . A 90 cm deep soil pedon was used for simulation with an assumed macropore volume fraction of 2.0% at the bottom. Based upon the variability of organic matter content with depth, the estimated macropore volume fraction at the soil surface was 3.14%.

The soil pedon was assumed to be solute free prior to simulation and the pressure distribution was hydrostatic with a linear variation of pressure from -150 cm at the land surface to -60 cm at the bottom of the simulated zone. The upper boundary condition is precipitation or sprinkler irrigation at  $6.25 \times 10^{-4}$  cm s<sup>-1</sup>, which is based on a Type II storm (US Department of Agriculture, 1986) and using the frequency distribution relationship for a two-year 24 h storm from Huff and Angel (1989). This flux is five times the saturated hydraulic conductivity of the matrix region. Details are presented in Ray (1994). During the course of simulation, after the surface of the matrix region becomes saturated and ponding commences, we switch the boundary condition from a flux to a Dirichlet type condition and then recompute flux entering the matrix region. The excess water flux (i.e. the difference between the imposed flux boundary condition and the quantity entering the matrix region) is diverted from the matrix region to the macropore region. Once the macropore region is saturated at the surface, the boundary condition for both regions becomes a Dirichlet type. In these simulations, we assumed instantaneous runoff and ignored the effects of surface ponding.

We followed the approach of Gerke and van Genuchten (1993a) to compute the fluid and solute mass transfer coefficients as functions of the shape and size of soil aggregates, and the reactive chemical molecular diffusion coefficient. We assumed that the soil was composed of a mixture of spherical and slab type aggregates with diameters or widths limited to 1.0 cm. This gave an estimated  $\alpha_w$  value equal to approximately 5 cm<sup>-2</sup>. The

molecular diffusion coefficient was assumed to be  $10^{-5} \text{ cm}^2 \text{ s}^{-1}$  which gave an  $\alpha_s$  value of  $1.2 \times 10^{-4} \text{ s}^{-1}$ . In addition, the longitudinal and transverse dispersivities were assumed to be 1.0 cm each.

A number of scenarios were considered by Ray (1994) and Ray et al. (1996) to evaluate the impact of agricultural management practices on flow and transport. These include the impact of row versus non-row crops, soil layers, furrow and sprinkler irrigation, tillage practices, and organic amendment. In this paper, we discuss only one scenario: the impact of organic amendment in conjunction with two tillage practices for the transport of atrazine.

Wastes from feedlot operations and animal housing facilities are often applied to the land for disposal purposes and to increase the nitrogen content of soil. In addition, straw, leaves and other residues from growing crops are applied to land. When spread at the land surface, these materials can reduce water evaporation and water erosion (Brady, 1974). Such materials also increase soil organic matter content at depth, depending on whether the materials are surface applied or mixed into the soil with tillage. In this section, we evaluate the impact of increasing soil organic matter on flow and transport in soils under conventional and conservation tillage for nonrow crops.

The same 90 cm deep soil profile, with the previously described hydrostatic initial pressure distribution and hydraulic properties, macropore fractions, mass transfer coefficients, and rainfall boundary conditions was used. The one-dimensional model was employed since we simulated only non-row crops. We focused on the impact of conventional tillage versus conservation tillage in conjunction with organic amendment. We also considered a case with organic matter addition, but no tillage. In conservation tillage, we assumed that all the organic addition was retained in the top 7.5 cm. The organic matter content of the soil in the top 7.5 cm was increased from an initial average of 7–10%. For conventional tillage, we assumed the added organic matter was thoroughly mixed in the plow layer (top 25 cm). Before adding the organic matter, the composite (mixed) organic matter content for the top 25 cm was 5.0% and the addition of the organic matter increased the average organic matter content in this top 25 cm horizon to 5.9%.

We simulated the movement of atrazine in the soil column, using the  $K_{oc}$  value given in Jury et al. (1987): water solubility =  $70 \text{ mg L}^{-1}$ ,  $K_{oc} = 160 \text{ mL g}^{-1}$ , and  $K_d$  at 1% organic carbon =  $1.6 \text{ mL g}^{-1}$ . (One can estimate the organic carbon content from organic matter content by dividing the latter with 1.7; see Brady, 1974). We estimated  $K_d$  values for each region from the fraction of organic carbon and the organic carbon partition coefficient using the approach of Griffin and Roy (1985) and Hassett et al. (1983). Further, we assumed that the volume fraction of macropores in the soil was not influenced by the addition of the organic matter prior to planting. We used two reaction rates ( $k_m = k_f = 10^{-4} \text{ s}^{-1}$  and  $k_m = k_f = 10^{-3} \text{ s}^{-1}$ ) for the two regions, and these rates can be labelled as moderate to moderately fast as presented in Ray (1994).

The simulation was performed using an initial time step of 0.1 s and the total simulation time was 1000 s. The concentration profiles for atrazine with and without organic amendment, and either with conservation or conventional tillage are shown in Fig. 10. Although 1000 s is a relatively short simulation, the differences between the treatments are readily apparent. As expected, the liquid-phase concentration at a given depth in the macropore region for both chemicals is lowest for the soil with no tillage. Also, the concentration

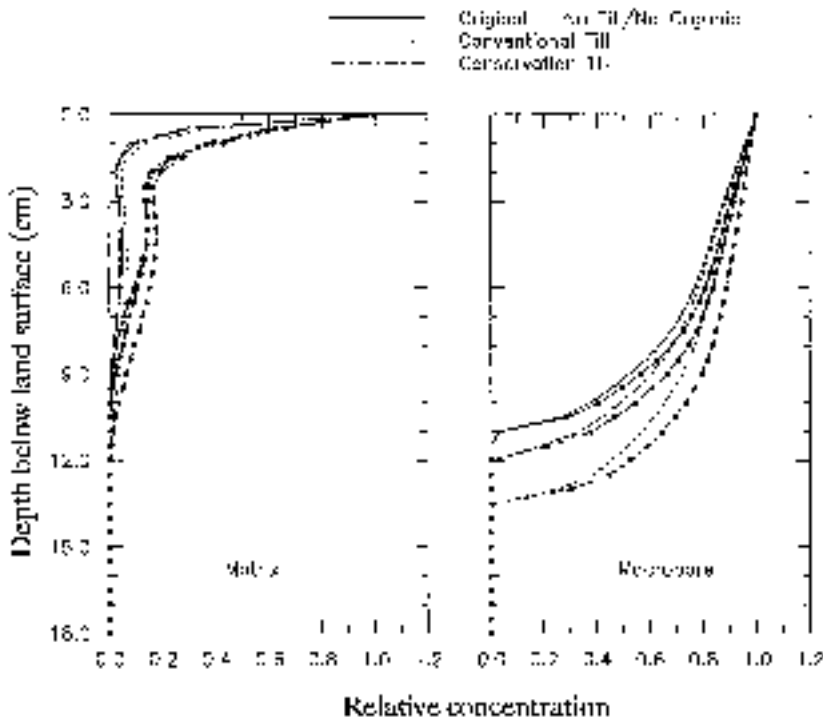


Fig. 10. Relative concentration profile for the matrix and the macropore regions for organically amended soil when atrazine is applied at the land surface after 900 s of simulation. (Note: • symbols indicate concentrations for  $k_m = k_f = 0.001 \text{ s}^{-1}$ ; curves without • symbols are for  $k_m = k_f = 0.0001 \text{ s}^{-1}$ ).

profile for conservation tillage was shallower than that for the conventional tillage. For conservation tillage, although there was an increase in organic carbon content in the top 7.5 cm compared to the baseline condition, there was also an increase in the macropore volume fraction in the top 7.5 cm owing to the tillage. For conventional tillage with the same amount of organic matter addition, there was a reduction in organic matter content in the top 7.5 cm of the soil profile because of the mixing of this organic matter with the low organic content soil at greater depths. The solute concentration in the matrix region for atrazine was highest for conventional tillage. The atrazine present in the matrix between the depths of 3 and 12 cm was a result of transfer from the macropore region. A similar simulation for trifluralin using a reaction rate of  $0.0001 \text{ s}^{-1}$  in the matrix and the macropore regions showed the presence of this compound only in the top 1.0 cm of the matrix region owing to its strong sorption properties.

## 9. Summary and conclusion

The dual porosity model is conceptually simple and provides a flexible approach for simulating preferential movement of water and reactive chemicals in variably saturated

macroporous soils. The model assumes two coexistent continua with water flow and chemical movement occurring in each region. A mass-conservative approach was used in the formulation of the flow equation for each region. Two equations (one for each region) at each node point in the discretized numerical model were solved simultaneously using fast iterative solvers. An important aspect in the conceptual formulation of the model is that the macropore volume fractions are allowed to vary with space as well as time. As illustrated in example applications, this is useful for simulating the impact of agricultural management practices on flow and transport.

Stable solutions of the flow equations were obtained, even for large magnitudes of the inter-region fluid mass transfer coefficient ( $\alpha_w$ ). However, large  $\alpha_w$  values caused numerical “overshoot” problems in the concentration profiles in the macropore region. Several approaches were taken to correct this problem, the most successful being an element-averaged computation of the advective solute mass transfer coefficient (the term  $\Gamma_w$ ) in the finite element formulation.

The model allows the use of linear kinetic sorption reactions to study the relative mobility of chemicals with different sorption properties. For example, the model is also capable of simulating the effects of degradation and plant extraction of water (presented in Ray, 1994). Simulations with soil layering indicated that when the water in the macropore region reaches the layer interface, it will spread laterally in both the macropore and the matrix regions. This may lead to important downslope flow and transport on sloping land forms. The model was used to simulate the impact of cropping practices, furrow irrigation, tillage, and organic amendment on water flow and chemical migration in macroporous soils.

The biggest challenge remaining for applying this model to a given field site is obtaining estimates of realistic hydraulic and transport parameters. For this model to be useful as a research or regulatory tool, it is essential that such parameter values be available. Controlled laboratory or field experiments and appropriate estimation methods are required in order to obtain realistic model parameters.

## Acknowledgements

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